

Kinematics of Anterior Cruciate Ligament Ruptures in World Cup Alpine Skiing

2 Case Reports of the Slip-Catch Mechanism

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Background: Based on visual video analyses of 20 injury situations, the main mechanism of anterior cruciate ligament (ACL) injury in World Cup alpine skiing, termed the “slip-catch” mechanism, was identified. This situation is characterized by a common pattern in which the inside edge of the outer ski catches the snow surface while turning, forcing the knee into valgus and tibial internal rotation. To describe the exact joint kinematics at the time of injury, a more sophisticated approach is needed.

Purpose: To describe the knee and hip kinematics in 2 slip-catch situations utilizing a model-based image-matching (MBIM) technique.

Study Design: Descriptive laboratory study.

Methods: Two typical slip-catch situations in World Cup alpine skiing reported through the International Ski Federation (FIS) Injury Surveillance System were captured on video with several camera views and high video quality. The injury situations were analyzed using the MBIM technique to produce continuous measurements of knee and hip joint kinematics.

Results: Within 60 milliseconds, the knee flexion angle increased rapidly from 26° to 63° in case 1 and from 39° to 69° in case 2. In the same period, we observed a rapid increase in internal rotation of the tibia with a peak of 12° and 9°, respectively. The knee valgus angle changed less markedly in both cases. We also observed a rapid increase of hip flexion as well as substantial hip internal rotation.

Conclusion: Knee compression and knee internal rotation and abduction torque are important components of the injury mechanism in a slip-catch situation.

Clinical Relevance: Prevention efforts should focus on avoiding a forceful tibial internal rotation in combination with knee valgus.

Keywords: knee injury; anterior cruciate ligament; professional alpine skiing; injury mechanism; injury biomechanics; video analysis

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The risk of anterior cruciate ligament (ACL) injury in professional alpine skiing is high.^{7,27,31} We have recently described the mechanisms of ACL injury in World Cup alpine skiing as well as the events leading to these injuries, based on visual video analyses of 20 injury cases.^{2,3} The main mechanism, termed the “slip-catch” mechanism, is characterized by a pattern in which the skier is out of balance backward and/or inward while turning. The skier loses pressure on the outer ski, which then drifts away from the body’s center of mass. While trying to re-establish grip with the outer ski, the skier extends his or her leg. The inside edge of the outer ski abruptly catches the snow surface, forcing the nearly straight knee into flexion/compression, valgus, and tibial internal rotation.

It has previously been shown that visual video analysis is not ideal for estimating detailed joint kinematics from video recordings.¹⁷ Thus, a more sophisticated approach is needed. Krosshaug and Bahr¹⁶ developed and validated a model-based image-matching (MBIM) technique to

describe the kinematic characteristics of ACL injuries in sports more accurately. Using the MBIM technique, it is possible to produce continuous estimates of joint angles and positions and thereby estimate the time of injury by assessing abnormal joint configuration and sudden changes in joint angular motion.¹⁸

A detailed description of joint kinematics in injury situations is needed to improve our understanding of the injury mechanism.¹⁵ Continuous estimates of knee and hip joint angles, as well as a description of the time of injury, can give valuable information to develop targeted prevention measures to reduce the risk of ACL injuries. This information can be used in the development of sport-specific training methods, and in alpine skiing, it may be critical for assessing the potential for improvements in boot-binding systems. Thus, knowledge of the injury mechanisms in terms of the skiing situation, skier behavior, and joint kinematics that represent different domains can provide important clues to how injuries can be prevented.¹³ Kinematic data obtained from video recordings can also verify whether simulation studies (mathematical, cadaveric, or in vivo) actually correspond to what is seen in real injury situations.¹⁵

To describe the joint kinematics of ACL injuries in alpine skiing, utilizing the MBIM technique, at least 2 camera views in which the injured knee is shown clearly from different perspectives are needed for a good result.¹⁶ Unfortunately, injury video of this quality is rarely available. The objective of this study was to examine the detailed time course of knee and hip kinematics of 2 typical slip-catch situations in World Cup alpine skiing reported through the International Ski Federation (FIS) Injury Surveillance System.^{7,8}

MATERIALS AND METHODS

Injury Case 1

One of the 10 slip-catch situations included in the previous visual video analyses^{2,3} was captured from 2 nearly perpendicular camera views, lending itself to this approach. A 29-year-old male alpine skier suffered an ACL tear of the right knee in a World Cup slalom race in Alta Badia, Italy, on December 17, 2007. Before the injury situation, the racer was late in line after a change of rhythm in the set course. He skied too directly into the entrance of a hairpin while leaning backward and tried to manage the next gate with an inner ski turn. The injury situation is described in Figure 1 (see also Video 1, previously published as a supplement to Bere et al²; available at <http://ajsm.sagepub.com/content/39/7/1421/suppl/DC1>).

Injury Case 2

An ACL injury from the 2011-2012 season, also classified as a slip-catch situation, was captured with 4 camera views. A 26-year-old female alpine skier suffered an ACL tear of her left knee in a World Cup parallel slalom race in Schladming, Austria, on March 16, 2012. Before the

injury situation, the racer was late on the ideal line; thus, she had a more direct approach into the next gate in a right-hand turn. In the steering phase out of the fall line, she was leaning backward and inward, losing pressure on the outer ski. The injury situation is described in Figure 2 (see also Video 2, available in the online version of this article).

We obtained written informed consent from the athletes before the analyses as well as anthropometric data and measurements of the skis and boots.

Video Editing

Infront Sports & Media (Zug, Switzerland) provided video footage (analog BetaSP tape) of the first injury case. The analog file was digitized to a QuickTime (.mov) file (Episode Engine Admin v 5.0, Telestream Inc, Nevada City, California; Apple, Cupertino, California) with a DV25 PAL codec. Footage of the second case was obtained from Austrian Broadcasting Corporation (ORF, Wien, Austria) and received as a high-definition video file. For both cases, we deinterlaced 640 milliseconds of the video file to a tagged image file format (TIFF) using Adobe Premiere Pro (version 1.5, Adobe Systems Inc, San Jose, California) and Adobe Photoshop (version CS, Adobe Systems Inc) to increase the effective frame rate from 25 Hz to 50 Hz. In each case, manual synchronization between the different camera angles was performed using key events in each camera view (ie, ski's contact with snow and skier's contact with gate). The image sequences were then placed next to each other in a new video compilation and stored as an uncompressed audio video interleave (AVI) file using Adobe After Effects (version CS, Adobe Systems Inc).

MBIM Technique

We used the MBIM technique^{16,18} to estimate kinematic characteristics throughout the 640-millisecond video sequence for each of the ACL injuries. The matching process was performed using the commercially available program Poser 4 and the Poser Pro Pack (Curious Labs Inc, Santa Cruz, California). A model of the surroundings was built and manually matched to the background video picture for each frame in all camera views. Surrounding landmarks/reference points, such as gate poles and safety nets, were presented in the video sequences, but the exact coordinates of these landmarks were unknown. Thus, we were not able to accurately position the skier in space, but we used the matched landmarks as guidance to determine the camera positions.

We utilized a skeleton computer model from Zygote Media Group Inc (Provo, Utah) for the skier matching process. This model consisted of 21 rigid segments as well as skis and poles. The model had 57 degrees of freedom and was hierarchically structured, using the pelvis as the parent segment. Pelvic motion was described by 3 rotational and 3 translational degrees of freedom, and the motion of the remaining segments was then described with 3 rotational degrees of freedom relative to their parent, for example, the shank relative to the thigh.

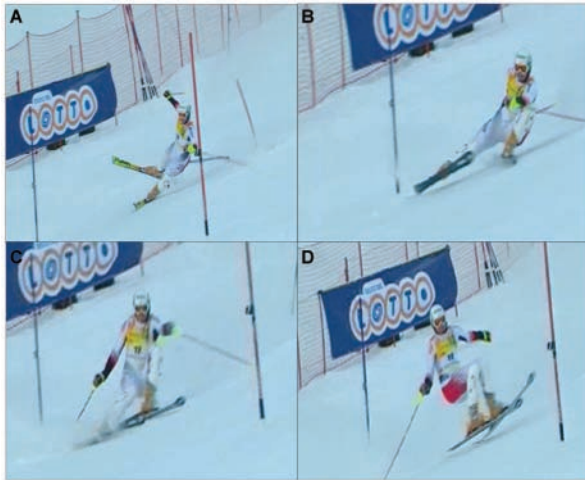


Figure 1. A slip-catch situation (anterior cruciate ligament injury to the right knee). (A) At –400 milliseconds, the skier is out of balance backward and inward in the steering phase out of the fall line. (B) At –120 milliseconds, as the skier tries to regain snow contact with the unweighted outer ski, he extends his right knee. (C) In the index frame, the outer ski catches the inside edge abruptly. (D) At +200 milliseconds, the skier falls backward to his right. (From Bere et al²; ©2011 by American Orthopaedic Society for Sports Medicine, reprinted with permission.)

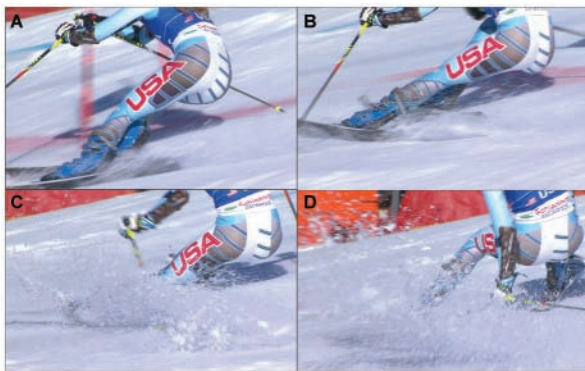


Figure 2. A slip-catch situation (anterior cruciate ligament injury to the left knee). (A) At –300 milliseconds, the skier is out of balance backward and inward in a right-hand turn. (B) At –160 milliseconds, she loses pressure on the outer ski, which drifts away from the body's center of mass. (C) In the index frame, the outer ski catches the inside edge abruptly. (D) At +140 milliseconds, the skier falls backward to her left.

The dimensions of the skeleton model were based on direct anthropometric measurements of the injured skier. The skis were assumed to be rigidly connected to the feet. Tibial rotation was matched using the orientation of the ski boot as guidance, and the rotation was fully assigned to the knee, as the ski boot was assumed to prevent ankle rotation.^{1,4,5,28} In addition, the skin-tight racing

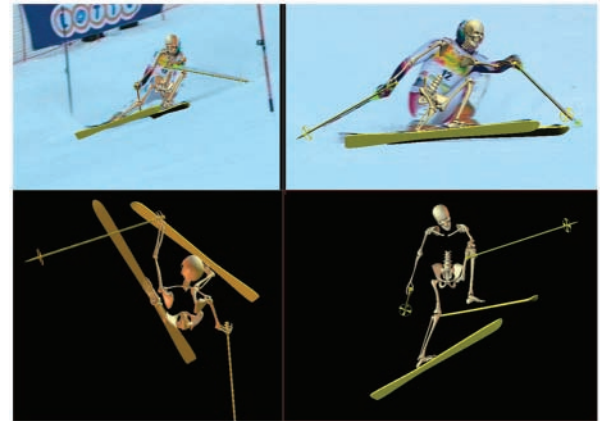


Figure 3. Matching process of the injury situation (case 1), corresponding to 460 milliseconds into the video sequence. The 2 top images show the customized skeleton model matched to the skier in the background video pictures, while the bottom images show the skeleton model from alternative views generated in Poser.

suit had reference points that were used to assess the body configuration, such as thigh and shank rotation. The frame-by-frame matching process has been described in detail in previous studies.^{16,18}

A frame of the matched video sequence is shown in Figure 3 (see also Video 2, available online). To minimize the bias resulting from single-operator judgment, 3 experts gave their opinion on the matching results. The matching process was then adjusted accordingly until a consensus was reached. The time histories of the joint angle were read into Matlab (MathWorks Inc, Natick, Massachusetts) with a customized script for data processing. Hip and knee kinematics were deduced by the joint coordinate system described by Grood and Suntay.¹¹

RESULTS

Injury Case 1

We analyzed 640 milliseconds of the injury situation, and the outer ski was airborne until 420 milliseconds into this video sequence. At that moment, the skier regained snow contact with the ski tail on a slightly flexed knee, as shown in Figure 4A. Within the next 60 milliseconds, while the ski tail appeared to become more loaded (more visible snow spray), we observed that the tibial internal rotation and knee valgus angle increased slightly, while the knee flexion angle stayed nearly constant. Then, when the midportion of the ski engaged the snow surface (Figure 4B), the knee flexion angle increased rapidly from 26° to 63° within 60 milliseconds. In the same period, we observed a rapid increase of tibial internal rotation with a peak of 12°. The knee valgus angle increased more gradually with a peak of 13°, which occurred 80 milliseconds after the peak tibial internal rotation. We also observed that the hip flexion angle increased

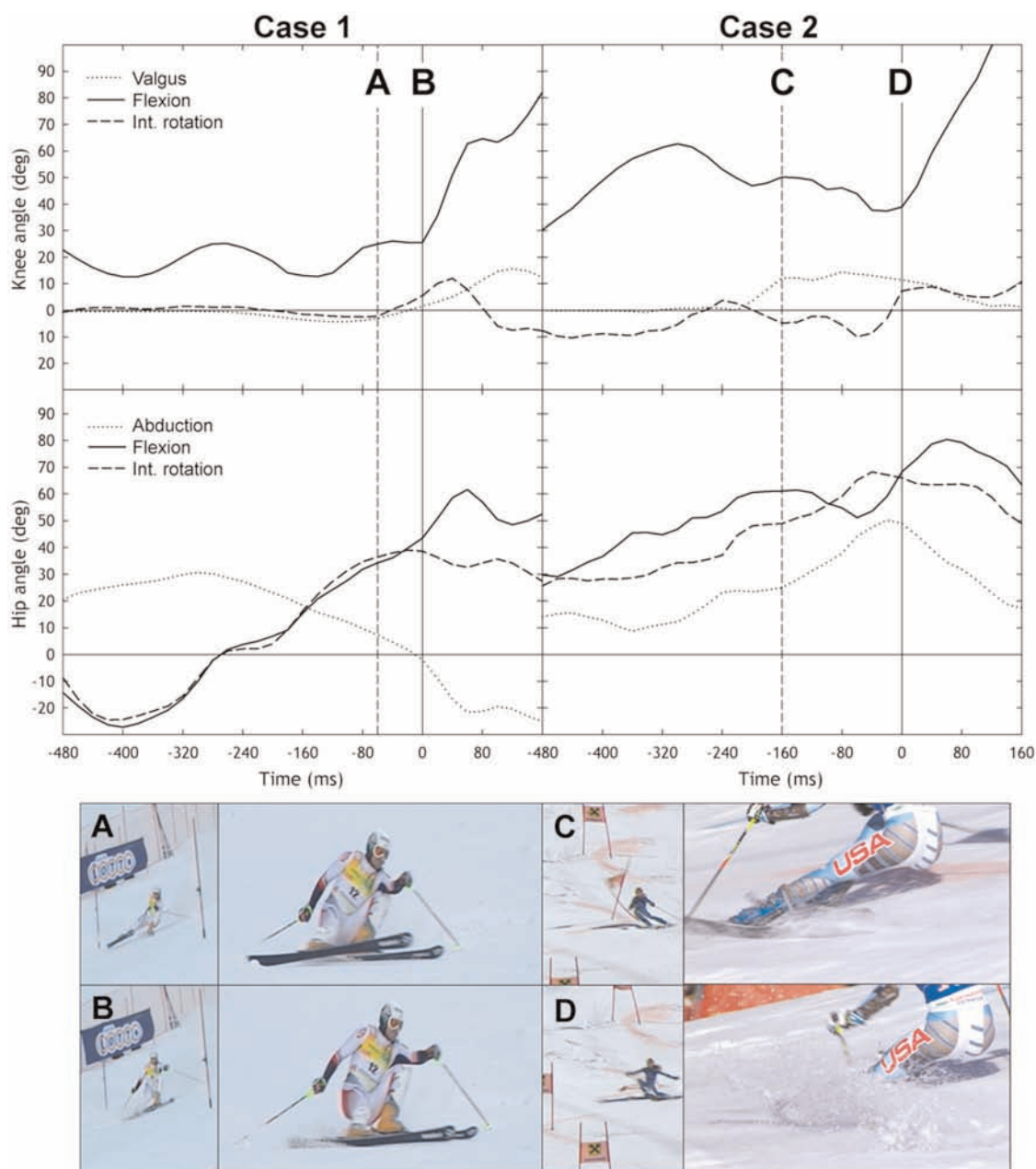


Figure 4. Estimated knee and hip joint angles (deg) of the injured knee (outer leg) throughout the 640-millisecond matched video sequence. Case 1: The dotted vertical line, A, indicates when the ski tail touched the snow surface (panel A), while the solid line, B, indicates the onset of injury (time 0), 60 milliseconds later (panel B). Case 2: The dotted vertical line, C, indicates a sudden increase in the knee valgus angle (panel C), while the solid line, D, indicates the onset of injury (time 0), 160 milliseconds later (panel D).

abruptly and simultaneously with the rapid increase in knee flexion, while the hip internal rotation stayed nearly constant at approximately 35° .

Injury Case 2

We observed that when the outer ski drifted away from the body's center of mass in a right-hand turn, the valgus angle of the left knee increased suddenly from 0° to 12° within 60

milliseconds (Figure 4C). During the next 120 milliseconds, the valgus angle stayed almost constant, while the outer ski was still drifting on the snow surface, trying to regain grip. Then, when the inside edge of the ski suddenly caught into the snow (Figure 4D), we observed that the knee flexion angle increased rapidly from 39° to 69° within 60 milliseconds. We also observed a rapid increase of tibial internal rotation with a peak of 9° in the same period. The knee valgus angle decreased slightly from a peak of 14° , observed

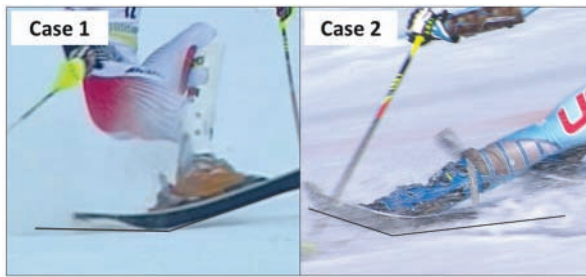


Figure 5. A carving ski will rotate inward because of the ski's self-steering effect when a load is applied on the inside edge of the ski. This can generate an internal tibial torque when the load is applied behind the projection of the tibial axis, and the skier is out of balance, unable to adjust to the rotary motion.

120 milliseconds before the peak tibial internal rotation. Similar to case 1, we also observed that the hip flexion angle increased abruptly and simultaneously with the rapid increase in knee flexion, while hip internal rotation did not change substantially. In case 2, the hip was internally rotated at approximately 65° .

DISCUSSION

We observed sudden changes in knee and hip joint angles, in both cases, when the inside edge of the outer ski caught the snow surface. Within 60 milliseconds, the knee flexion angle increased from 26° to 63° in case 1 and from 39° to 69° in case 2. In the same period, we observed a rapid increase in tibial internal rotation with a peak of 12° and 9° , respectively. The knee valgus angle changed more gradually in both cases. We also observed a rapid increase of hip flexion and substantial hip internal rotation.

Slip-Catch Kinematics

The leg was abruptly and forcefully compressed from a relatively straight position in both cases. This motion pattern indicates that there is likely a high compression force of the knee joint involved. Previous studies support that a combination of knee joint compression, tibial internal rotation, and knee valgus is an important component of the ACL injury mechanism in team sports such as handball, basketball, and football.^{12,13,21,23,24,26} A proposed hypothesis is that noncontact ACL injuries seem to occur approximately 40 milliseconds after initial contact and that knee abduction loading and lateral compression generate tibial internal rotation and anterior tibial translation due to the joint surface geometry and possibly quadriceps drawer, resulting in an ACL tear.¹³ Interestingly, the motion pattern observed in the current study closely resembles the motion pattern observed for noncontact ACL injuries. However, there are forces and moments involved in alpine skiing that make a slip-catch situation different from a noncontact ACL injury situation.

In a slip-catch situation, we have observed that the skier is out of balance backward/inward.² This will generate an

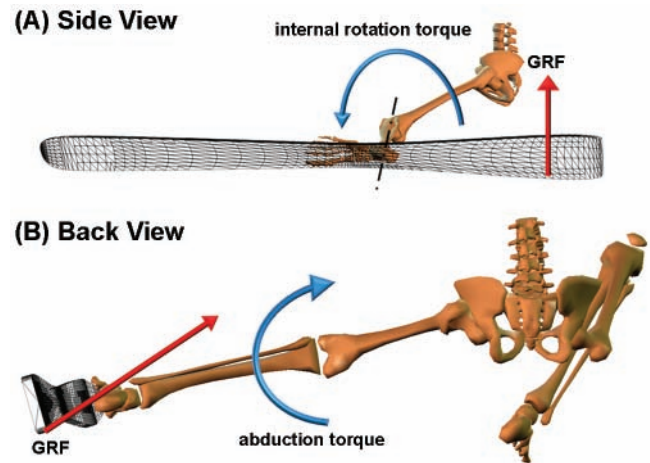


Figure 6. The assumed forces and moments involved in a slip-catch situation. The top image (A) shows a lateral load applied behind the projection of the tibial axis, which acts as a lever to internally rotate the tibia relative to the femur. The bottom image (B) shows how edging a carving ski may cause knee abduction loading. GRF, ground reaction force.

internal tibial torque by 2 mechanisms. First, a carving ski will rotate inward because of the ski's self-steering effect.²⁰ In a balanced position, the skier will be able to follow the path of the ski by using appropriate technical approaches,^{20,29} but when the skier is out of balance, he or she may be unable to adjust to the rotary motion. Thus, the self-steering effect of the ski can generate an internal tibial torque by forcing the ski to carve inward (Figure 5). Second, when a lateral load is applied behind the projection of the tibial axis, the ski will act as a lever to generate internal tibial torque.³⁰ A more backward lean will give a longer moment arm to the tibial axis and thus a larger internal rotation torque (Figure 6A). Although it is difficult to reliably measure small internal rotations of the knee using the MBIM method, the large hip internal rotation combined with pressure on the medial side of the rear ski makes this scenario very likely.

In addition, it has been reported that edging a carving ski while turning is associated with knee valgus angles up to 12° .^{4,10,33} An increased knee valgus angle will likely also contribute to an increased abduction torque (Figure 6B). Catching an inside edge, while being out of balance backward/inward, may therefore represent a serious hazard to the ACL.^{2,18,30}

It has been suggested that in a backward-bending posture while skiing, boot-induced anterior drawer⁶ can also strain the ACL.¹⁴ In addition, it has been proposed that a quadriceps contraction may cause tibial anterior drawer when the skier tries to recover from an unbalanced position backward on slightly flexed knees.^{9,22} Because of clothing and limited picture resolution, we were not able to evaluate whether tibial anterior drawer was present in these slip-catch situations. However, based on the sudden changes in joint angles observed with the MBIM technique, we suggest that the ACL injury was caused by simultaneous knee compression and knee internal rotation and abduction torque. It therefore seems likely that the

ACL ruptured shortly after the onset of large changes in knee flexion.

Methodological Considerations

This study has shown that we can use the MBIM technique to provide continuous measurements of knee and hip kinematics during 2 slip-catch situations representative of a series of similar injuries from World Cup alpine skiing. It is difficult to determine the accuracy of the current matching process, but the nearly perpendicular camera views and the high video quality made it possible to achieve a good match between the skier and the skeleton model for the video images in all frames. Even though this technique is subjective and dependent on the operator's ability to perform the model matching process consistently, the method is likely to be far more accurate than visual video analysis.¹⁶⁻¹⁸ An advantage of analyzing a skiing injury with this method is that tibial rotation relative to the femur can likely be matched more accurately than for team sports injuries based on the orientation of the ski and boot. The tibial rotation was fully assigned to the knee, as the stiff and tight ski boot was assumed to prevent ankle rotation. This assumption has not been validated but is supported by previous studies.^{1,4,5,28} We also used reference points on the skin-tight racing suit to determine rotation of the thigh and shank. Similar to traditional 3-dimensional motion analysis with skin markers, there are mainly 2 sources of error that may affect hip rotation, knee rotation, and valgus angle when utilizing the MBIM technique: (1) movement of the reference points on the racing suit relative to the underlying soft tissue, and (2) movement of the soft tissue relative to the underlying bone. Because of the material specifications of the racing suit,³² we expect minimal movement of the suit relative to the skin, but it is difficult to predict how the soft tissue moves with respect to the underlying bone. It has been recognized that soft tissue artifact is the most significant source of error in human movement analyses.^{19,25} Thus, we should keep in mind that the estimated kinematics will not be a true representation of the skeletal motion, even though the matching process is performed optimally. Three experts gave their opinion on the matching results, and the matching process was then adjusted iteratively until a consensus was reached. We performed sensitivity analyses of the matching process in which the femur was axially rotated inward and outward to ensure that we presented the best possible estimates of knee and hip joint angles. It would have been preferable to have more cases, but video of sufficient quality for this advanced analysis is rare: of all the ACL injury videos we collected from 6 World Cup seasons (2006-2012), only these 2 cases were suitable.

CONCLUSION

We were able to assess sudden changes of knee and hip angles for 2 slip-catch situations representative of a series of similar injuries from World Cup alpine skiing by using

the MBIM technique. This study supports previous visual video analyses,² suggesting that knee compression and knee internal rotation and abduction torque are important components of the injury mechanism in a slip-catch situation.

More than half of the ACL injuries in World Cup alpine skiing occur in a slip-catch situation, and to reduce the risk of these injuries, it is important to avoid a situation in which the knee is rapidly forced into internal rotation and valgus while slightly flexed. Even though a relatively precise time sequence of knee and hip angles was obtained for 2 representative slip-catch situations, we need to analyze several cases, utilizing the MBIM technique, to evaluate whether these estimates are general trends and to quantitatively describe the joint kinematics of the slip-catch mechanism.

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A Video Supplement for this article is available in the online version or at <http://ajsm.sagepub.com/supplemental>.

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